

SETS — Short Revision Notes

Remember Level: Recall Facts & Basic Knowledge

1. Key Definitions

Term	Definition	Example
Set	A collection of well-defined objects	$\{1, 2, 3, 4, 5\}$
Elements	The objects / members in a set	1, 2, 3, 4, 5
$\in$ (Belongs to)	Symbol meaning "is an element of"	$2 \in \{1, 2, 3\}$
$\notin$ (Does not belong to)	Symbol meaning "is not an element of"	$7 \notin \{1, 2, 3\}$
$n(A)$	Number of elements in set A	$A=\{2,4,6\}$, $n(A)=3$
ϕ (Null Set)	A set with no elements	$\phi = \{ \}$
ε (Universal Set)	All elements under consideration	$\varepsilon = \{1,2,\dots,10\}$

2. Types of Sets

Type	Definition	Example
Finite Set	Definite / countable number of elements	$\{2,4,6,8,10\}$ (5 elements)
Infinite Set	Endless number of elements	$\{1,2,3,4,5,\dots\}$
Null Set (Empty)	Has no elements	ϕ or $\{ \}$
Singleton Set	Has exactly one element	$\{5\}$
Equal Sets	Exactly the same elements	$A=\{1,2,3\}$, $B=\{1,2,3\} \rightarrow A=B$
Equivalent Sets	Same number of elements	$A=\{1,2,3\}$, $B=\{4,5,6\} \rightarrow A \sim B$

3. Set Notation

Symbol	Meaning	Example
$\{ \}$	Curly brackets — enclose elements	$\{2,4,6\}$
$\in$	Belongs to / is an element of	$2 \in \{2,4,6\}$
$\notin$	Does not belong to	$5 \notin \{2,4,6\}$
$n(A)$	Number of elements in set A	$n(\{2,4,6\}) = 3$
ϕ	Null set (empty set)	$\phi = \{ \}$
$\subset$	Subset of	$\{2,4\} \subset \{1,2,3,4\}$
$\not\subset$	Not a subset of	$\{2,5\} \not\subset \{1,2,3,4\}$
ε (or U)	Universal set	$\varepsilon = \{1,2,3,4,5\}$
A' (or A^c)	Complement of set A	$\varepsilon=\{1..5\}$, $A=\{2,4\}$, $A'=\{1,3,5\}$
$\cap$	Intersection (common to both)	$A \cap B$
$\cup$	Union (all elements in both)	$A \cup B$
$A \sim B$	Equivalent sets ($n(A)=n(B)$)	$A \sim B$

4. Set Operations

Operation	Symbol	Definition	Example
Intersection	$\cap$	Elements common to both sets	$A=\{2,4,6\}, B=\{4,6,8\} \rightarrow A \cap B=\{4,6\}$
Union	$\cup$	All elements in both sets	$A=\{2,4,6\}, B=\{4,6,8\} \rightarrow A \cup B=\{2,4,6,8\}$
Complement	' (prime)	In universal set but not in A	$\varepsilon=\{1..7\}, A=\{2,4,6\} \rightarrow A'=\{1,3,5,7\}$
Subset	$\subset$	All elements of one set are in another	$\{2,4\} \subset \{2,4,6\}$
Disjoint	$A \cap B = \phi$	Sets with no common elements	$A=\{1,3,5\}, B=\{2,4,6\}$

5. Key Relationships

Relationship	Formula	Example
Equal Sets	$A = B$ (same elements)	$A=\{1,2,3\}, B=\{1,2,3\}$
Equivalent Sets	$n(A) = n(B)$	$A=\{1,2,3\}, B=\{4,5,6\}$
Subset	$A \subset B$	$\{2,4\} \subset \{2,4,6,8\}$
Disjoint Sets	$A \cap B = \phi$	$A=\{1,3,5\}, B=\{2,4,6\}$
Complement	$A \cup A' = \varepsilon$	$A=\{2,4,6\}, A'=\{1,3,5,7\}$
Intersection	$A \cap B = B \cap A$	$\{2,4\} \cap \{4,6\} = \{4\}$
Union	$A \cup B = B \cup A$	$\{2,4\} \cup \{4,6\} = \{2,4,6\}$

6. Venn Diagram Symbols

Symbol	Meaning	Venn Diagram
ε	Universal Set	Rectangle
A	Set A	Circle / ellipse inside rectangle
A'	Complement of A	Outside the circle, inside rectangle
$A \cap B$	Intersection of A and B	Overlapping region of two circles
$A \cup B$	Union of A and B	All parts of both circles
ϕ	Disjoint sets	Two non-overlapping circles

7. Quick Reference Table

Concept	Symbol	Example
Element	$\in$	$3 \in \{1,2,3\}$
Not an element	$\notin$	$5 \notin \{1,2,3\}$
Number of elements	$n(A)$	$n(\{1,2,3\}) = 3$
Null set	ϕ or $\{ \}$	$\{ \}$
Universal set	ε	$\{1,2,3,4,5\}$
Subset	$\subset$	$\{1,2\} \subset \{1,2,3\}$
Intersection	$\cap$	$\{1,2\} \cap \{2,3\} = \{2\}$
Union	$\cup$	$\{1,2\} \cup \{2,3\} = \{1,2,3\}$
Complement	A'	$A=\{1,2\}, A'=\{3,4\}$

8. Check Yourself — Quick Questions

No.	Question	Answer
1	What is a set?	A well-defined collection of objects
2	Symbol for "belongs to"?	$\in$
3	Symbol for "does not belong to"?	$\notin$
4	What does $n(A)$ mean?	Number of elements in set A
5	What is a null set?	A set with no elements
6	What is a universal set?	Set of all elements under consideration
7	Symbol for null set?	ϕ or $\{ \}$
8	What is a subset?	A set whose elements are all in another
9	Symbol for subset?	$\subset$
10	Symbol for intersection?	$\cap$
11	Symbol for union?	$\cup$
12	What is a finite set?	A set with a definite number of elements
13	What is an infinite set?	A set with an endless number of elements
14	What are equal sets?	Sets with exactly the same elements
15	What are equivalent sets?	Sets with the same number of elements
16	What is a complement of a set?	Elements in universal set not in the set
17	Symbol for complement?	A'
18	What are disjoint sets?	Sets with no common elements

9. Quick Memory Aids

Vocabulary

SET = Collection of well-defined objects

ELEMENTS = Objects in a set

Symbols

in / belongs to element is in the set

not in element is not in the set

{ } "the set of"

$n(A)$ number of elements in A

null set empty set (no elements)

universal set all elements considered

subset every element is inside another set

intersection AND = common elements

union OR = all elements together

Types of Sets

FINITE = countable number of elements

INFINITE = endless number of elements

NULL = no elements

EQUAL = same elements

EQUIVALENT = same number of elements

Venn Diagram

Rectangle Universal Set

Circle Set A

Overlapping region Intersection (A and B)

All circles together .. Union (A or B)

Outside the circle Complement (A')

Revision Tip: Memorize all set symbols and definitions. Practice writing sets in different ways — descriptive, roster, and Venn diagram. Understanding the symbols is the first step to understanding sets.